

ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification

REPORT - WEEK 3

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GitHub Repository:

https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

1) Work done this week

→ The Germany and Uruguay CSV datasheets were loaded and inspected using Python and Pandas and Jupyter. The initial analysis focused on understanding the dataset structure, columns names, data types, and available features.

→ **Feature Analysis:** The available feature were examined, including Sentinel-2 optical bonds (B01-B12), soil properties, temperature, precipitation, and terrain-related information. The dataset structure was also checked to understand the available samples and time steps.

→ **Missing-Value Analysis:** Missing values were identified in the times and sample columns. Further investigation is required to understand their causes and determine appropriate preprocessing methods.

→ **Initial SAR Data Investigation:** The available CSV bands names were checked for Sentinel-1 VV and VH polarization data. These bands were not clearly identified in the inspected files, so separate SAR data extraction through Google Earth Engine will be investigated.

2) Work Planned for Next Week

- Verify the meaning of the target variable and crop labels.
- Calculate NDVI, NDRE, NDMI, and EVI using the appropriate optical bands.
- Generate NDVI time-series graphs for Germany and Uruguay.
- Extract Sentinel-1 VV/VH data and analyse its temporal patterns.
- Select a relevant research paper and prepare notes from 4–5 related studies.
- Develop a preliminary Random Forest baseline classifier, subject to label verification, and record the results.

3) Challenges Faced

→ During this week, we encountered the following challenges:

- **Data understanding:** Understanding the meaning and organization of the available bands, time steps, crop codes, and target values required further investigation.
- **Missing values:** Some columns contain missing values, so their effect on subsequent analysis must be evaluated.
- **SAR data availability:** The expected Sentinel-1 VV and VH bands were not found in the inspected CSC files. Additional SAR data extraction through Google Earth Engine is therefore required.
- **Optical-index calculation:** Calculating NDVI, NDRE, NDMI, and EVI requires selecting the correct spectral bands and ensuring that the input data are suitable for each formula.
- **Classification target:** The crop labels and target values must be verified before selecting a valid classification target.

- **Research and implementation:** Selecting a suitable base paper, reviewing related literature and developing an initial baseline model remain important tasks.

4) References:

→ Google Earth Engine – Sentinel-1 SAR GRD.

Documentation describing the Sentinel-1 Ground Range Detected dataset and its available radar polarization bands.

https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD

→ Google Earth Engine – Sentinel-2 Surface Reflectance Harmonized.

Documentation describing Sentinel-2 surface reflectance data and the available optical bands.

https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

→ A. Miranda et al., "YieldSAT: A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction," CVPR 2026. Project page: <https://yieldsat.github.io/>

→ Studies on Sentinel-1/Sentinel-2 fusion for crop-type mapping in Germany and South America